Project Design Phase-II

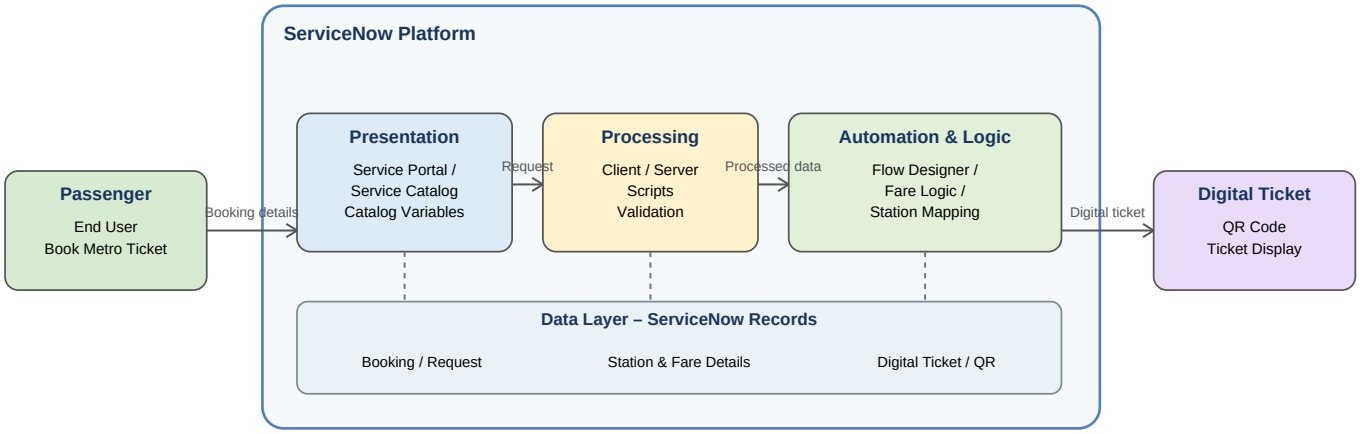
Technology Stack (Architecture & Stack)

Date	6 September 2026
Team ID	Individual
Name	Jonnalagadda Sai Sri Poornima Lahari
Project Name	Digital Metro Ticket Booking System using ServiceNow
Maximum Marks	4 Marks

Technical Architecture:

The proposed solution uses ServiceNow as the application platform for digital metro ticket booking. Passengers submit a booking through a Service Catalog item by providing source, destination, passenger type, and number of tickets. The system processes the booking, maps station details, calculates the fare, and generates a digital ticket with a QR code. The ticket can then be displayed through the configured ServiceNow user interface.

Architecture Flow



Core flow: Booking → Validation → Fare & Station Mapping → Ticket → QR Code

Optional digital payment channels can be integrated separately; no specific payment API is defined in the project details.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Interface through which passengers submit and view metro ticket booking details.	ServiceNow Service Portal / Service Catalog
2	Application Logic-1	Captures source, destination, passenger type, and number of tickets through the booking form.	Service Catalog Item, Catalog Variables
3	Application Logic-2	Validates booking inputs and supports dynamic form behavior.	Catalog Client Scripts / Client-side validation
4	Application Logic-3	Processes fare calculation and station mapping after the booking details are submitted.	ServiceNow Server-side Script / Business Logic
5	Automation	Coordinates ticket processing, generation, and status updates.	ServiceNow Flow Designer
6	Database / Records	Stores booking, station/fare, and digital ticket information required by the workflow.	ServiceNow platform tables / custom records
7	QR Code Integration	Generates or integrates a QR code containing the required digital ticket information.	QR Code generation / integration component
8	External API / Payment	Digital payment methods are part of the project objective, but a specific external payment API is not defined in the provided project details.	Not specified
9	Analytics / Reporting	Digital booking data can support analysis of passenger usage, peak periods, and operational patterns.	ServiceNow reporting / data analysis
10	Machine Learning Model	No machine-learning model is specified for the core metro ticket booking workflow.	Not applicable
11	Infrastructure (Server / Cloud)	Application, catalog, workflow, and records are hosted and executed on the ServiceNow instance.	ServiceNow Personal Developer Instance / Enterprise Instance

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	ServiceNow Configuration	The solution is configured within the ServiceNow platform using Service Catalog, scripting, workflow, and UI components.	ServiceNow platform configuration
2	Security	Access to booking records and administrative functions can be controlled using ServiceNow roles, permissions, authentication, and authorization.	ServiceNow roles, ACLs, authentication and authorization
3	Scalable Architecture	The platform-based solution can support additional metro stations, fare rules, passenger categories, ticket options, and higher booking volumes.	ServiceNow platform and Flow Designer
4	Availability	The application runs on the ServiceNow instance, allowing configured users to access the digital booking service.	ServiceNow instance
5	Performance	Automation reduces manual ticketing effort by collecting required variables, calculating fares, generating QR tickets, and maintaining booking records.	Service Catalog, Scripts and Flow Designer